

Dr. Vahram Martirosyan

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Abu Dhabi, UAE

Faculty Application — Ocean University of China (中国海洋大学), Qingdao
Interpretable Machine Learning for Ocean Science · HPC & Compiler Engineering · Explainable AI

Profile & Research Summary

Ph.D. in Computational Mathematics with **20+ years** spanning academic research, university teaching, and industrial software engineering. My research programme develops **interpretable machine-learning models for ocean and climate science**: stagewise-regularized polynomial regression ensembles that reconstruct global ocean temperature, salinity, and dissolved-oxygen fields and global significant wave height with accuracy competitive with tree ensembles and deep learning — but with explicit physical interpretability, microsecond-to-millisecond inference, and kilobyte-to-sub-megabyte footprints suitable for **on-board, edge, and autonomous deployment at sea**.

Complementary expertise in **high-performance computing and compiler engineering**: LLVM-based automatic parallelization (hpc++) and binary lifting of legacy numerical codes (LEAP), directly applicable to ocean-model acceleration and marine big-data processing. Broader research spans explainable AI for medicine (multi-class ECG), materials science, remote sensing, and precision agriculture, with **25+ publications** (one published IEEE-indexed conference paper, several under review) and a pending high-entropy-alloy design patent. 15+ years of university teaching (programming, systems, architecture, algorithms) with a track record of integrating research into the classroom. Currently **Assistant Professor and Course Team Leader** at Higher Colleges of Technology, Abu Dhabi, and **Chief Scientist** at ExplainableAI.app.

Research Interests

- **Interpretable ML for ocean science**: polynomial zone ensembles for global temperature, salinity, and dissolved-oxygen profile reconstruction; transparent alternatives to deep learning for ocean climatology.
- **Marine climate & wave modelling**: interpretable surrogates for global significant-wave-height climatology; coastal buoy transfer and uncertainty analysis.
- **Marine data science & edge deployment**: resource-constrained, latency-critical models (microsecond-to-millisecond inference, kilobyte-to-sub-megabyte memory) with int8/int16 quantization for at-sea, on-board, and autonomous platforms; spatial-block and profile-stratified validation.
- **HPC for ocean & climate computing**: LLVM-based automatic parallelization (hpc++), binary lifting of legacy numerical codes (LEAP), heterogeneous CPU–GPU execution.
- **Remote sensing data analysis**: multi-temporal optical and SAR feature engineering; oil-spill detection, crop-type mapping, and land/ocean-surface analysis.
- **Explainable AI (PRMs)**: polynomial regression models with fully readable coefficients for medicine, materials science, and agriculture.

Education

Ph.D., Computational Mathematics

2006 – 2009

Yerevan State University

Dissertation: “Construction of an Algorithm for the Pseudoinversion of Bidiagonal Matrices and an Investigation of its Parallelization”

M.Sc., Informatics and Applied Mathematics

2004 – 2006

Yerevan State University

B.Sc., Informatics and Applied Mathematics

2000 – 2004

Yerevan State University

Academic & Teaching Experience

Assistant Professor, Computer Information Science

Jan 2024 – Present

Higher Colleges of Technology, Abu Dhabi, UAE

- Courses: Fundamentals of Programming (Java), System Architecture & Integration, IoT & Security, Data Structures & Algorithms, Applied Discrete Mathematics, Computational Thinking, Statistics & Probability.

Course Team Leader

Jan 2025 – Present

Higher Colleges of Technology, Abu Dhabi, UAE

- Managing a multi-campus team of lecturers for System Architecture & Integration; developing course materials, assessments, and standardised delivery frameworks.

Senior Lecturer (Contract)

Feb 2021 – Dec 2023

Yerevan State University, Faculty of Informatics & Applied Mathematics

- Courses: Operating Systems. Supervised undergraduate research projects.

Faculty Lecturer (Contract)

Jan 2023 – Dec 2023

Université française en Arménie (UFAR), Yerevan

- Courses: Computer Architecture I, C Programming.

Visiting Lecturer (Contract)

Aug 2022 – Jan 2023

American University of Armenia, Yerevan

- Course: Introduction to Computer Science.

Informatics Teacher (Contract)

Sep 2023 – Dec 2023

PhysMath School After A. Shahinyan, Yerevan, Armenia

- Taught Informatics at a specialist physics-mathematics secondary school.

Founder & Lecturer

Jan 2019 – Jun 2023

MutEx School, Yerevan

- Founded and ran an independent programming school; designed curricula and delivered public lectures.

Lecturer

Sep 2013 – Dec 2019

Armsoft CS Learning Center, Yerevan

- Course: Operating Systems.

Lecturer

Feb 2015 – Jun 2015

Armenian Code Academy, Yerevan

- Course: .NET / C# Programming.

Senior Lecturer

Feb 2007 – 2014

Russian-Armenian (Slavonic) University, Yerevan

- Courses: Linux/POSIX Programming, Win32 API, Linux Device Driver Development, GUI Software Development (GTK+), Operating Systems, System Programming, Database Management Systems.

Courses Taught

- **Programming:** Fundamentals of Programming, C Programming, C++ Programming, High-performance computing in C++, .NET/C# Programming, Intro to Computer Science.
- **Systems & Architecture:** System Architecture & Integration, Computer Architecture I, Operating Systems, Linux/POSIX Programming, Win32 API Programming, Linux Device Driver Development.
- **Data & Mathematics:** Data Structures & Algorithms, Discrete Mathematics, Applied Discrete Mathematics, Database Management Systems, Statistics & Probability.
- **Emerging Topics:** IoT & Security, Computational Thinking, GUI Development (GTK+), System Programming.

Research & Industry Experience

Research

Chief Scientist

Jan 2026 – Present

[ExplainableAI.app](#)

- Directing scientific strategy for a no-code ML platform built entirely on intrinsically interpretable polynomial regression models — no black-box neural networks.
- Platform combines browser-side WebAssembly training (data never leaves the device), a hand-written C++ inference engine with sub-millisecond latency, and an AI App Builder that turns any trained model into a shareable live web app.

Chief Executive Officer

Jun 2022 – Jul 2023

System View Armenia, Yerevan

- Developed LLVM/Clang-based auto-vectoriser for Xilinx ACAP (Visual System Integrator toolchain).

Head of System Programming Laboratory

Jan 2009 – Jul 2013

Russian-Armenian University / ISP RAS, Yerevan

- Led RAS System Programming Lab; directed Linux Foundation verification of system libraries and kernel modules.

Junior Research Scientist

2003 – 2005

Institute of Mathematics, National Academy of Sciences of RA

- Parallelisation of numerical algorithms for integral equations using MPI.

Industry (Selected)

Director of Software Engineering

Sep – Dec 2023

Plat.AI, Yerevan

Associate Software Development Manager <i>HelpSystems Armenia, Yerevan</i>	Oct 2021 – Jun 2022
Principal Software Engineer / Team Lead <i>HelpSystems Armenia, Yerevan</i>	Feb 2017 – Oct 2021
Software Architect (Contract) <i>Priotix, Yerevan · smart-home IoT system</i>	2018 – 2020
Principal / Senior Software Developer <i>Optym, Yerevan · logistics optimisation</i>	Aug 2015 – Feb 2017
Senior Software Developer <i>ArdInnoTekh LLC, Yerevan</i>	Aug 2013 – Jul 2015
Software Developer <i>ISMOTech, Yerevan · firmware, C++/C#, ASP.NET</i>	Oct 2006 – Dec 2008
Software Developer <i>Lycos Europe · PHP/MySQL</i>	Mar 2005 – Oct 2006

Publications

Ocean & Climate Data Science

- **Martirosyan, V.** “On-Device TinyML for Multi-Variable Ocean Profiling.” *Under review, eICocean 2027* (IEEE-indexed), 2026.
($R^2 = 0.973$ temperature / 0.794 salinity / 0.895 dissolved oxygen; 4.4 μ s desktop inference; 1.09 ms on ESP32 and 1.49 ms on RP2350; 180 μ J/inference; 3–4 orders of magnitude smaller and faster than tree ensembles.)
- **Martirosyan, V.** “Interpretable Polynomial Zone Ensembles for Global Ocean Temperature and Salinity Profile Reconstruction.” *Under review* (Q1 journal) and *prepared for IEEE JSTARS*, 2026.
(WOA 2018: 534,090 vertical samples, 31,596 profiles; pooled $R^2 = 0.9703$ temperature / 0.8376 salinity; profile-stratified & spatial-block CV vs. Random Forest, XGBoost, ridge; sub-millisecond inference, sub-megabyte footprint; learned terms recover known oceanographic structure.)
- **Martirosyan, V.** “Depth-Zone Polynomial Regression for Global Ocean Climatology.” *Under review* (IEEE journal), 2026.
(IAP T/S Climatology 2024; elastic-net degree-6 models per zone: mixed layer, thermocline, deep ocean; spatially adaptive subsampling 534K $\rightarrow$ 177K rows.)
- **Martirosyan, V.** “Auditable Kilobyte-Scale Surrogates for the Global Significant-Wave-Height Climatology.” *Prepared for IEEE JSTARS*, 2026.
(Global wave reanalysis; $R^2 = 0.945$, RMSE 0.262 m; cell-holdout $R^2 = 0.933$; region-holdout $R^2 = 0.73 \pm 0.18$; coastal R^2 improves 0.34 $\rightarrow$ 0.52 with in-situ buoys; degree-8, 3,002-term model.)
- **Martirosyan, V.** “Portability and Limits of an Interpretable Polynomial Framework: A Global Significant-Wave-Height Case Study.” *Working paper*, 2026.
(Anomaly $R^2 = 0.875$; offshore buoy RMSE ≈ 0.5 m; negative result reported for a distance-to-coast feature.)
- **Martirosyan, V.** “Deep-Ocean Temperature Profile Reconstruction via Interpretable Polynomial Zone Ensembles.” *Working paper*, 2026.
(Stagewise-Rank-Polynomial-Regression; 5,005-term basis; 2.6 μ s per-profile inference in C++; edge-deployment analysis.)

- **Martirosyan, V.** “Global Ocean Salinity Profile Reconstruction with Interpretable Polynomial Zone Ensembles.” *Working paper*, 2026.
(WOA dataset; degree-8 expansion, 2,001 coefficients per zone; 23 μ s inference; river-mouth salinity correction; residual ensemble.)
- **Martirosyan, V.** “Blended Ocean Visualization: Combining Sparse Polynomial Regression with Residual Fields.” *Working paper*, 2026.

High-Performance Computing & Compilers

- **Martirosyan, V.** “LEAP: Lifting Legacy x86-64 Binaries to Reconfigurable IR for Automatic Parallelization.” *Submitted to Frontiers of Computer Science*, 2026.
(43-program liftability study: 100% at -O0, 98% at -O1/-O2/-O3; 21-benchmark suite; $3.7\times$ geom. speedup from source IR on 8-core CPU; uplifted binary path $10.3\times$ geom. over -O0, $2.1\times$ faster than -O2; GPU offload $246\times$; all 7 NAS Parallel Benchmarks lifted, parallelized, and verified byte-identically; 377/377 test cases pass.)
- **Martirosyan, V.** “hpc++: An LLVM-Based Automatic Parallelization System for C++ with GPU Acceleration.” *arXiv preprint*, Dec 2025.
(Seven parallelization strategies; zero source modification; 24 benchmarks; $247\times$ peak GPU speedup, $97\times$ peak CPU, $16.8\times$ average; 100% numerical correctness.)

Explainable AI: Medicine, Materials & Remote Sensing

- **Martirosyan, V.** “Stagewise Regularized Polynomial Regression: A Hybrid Optimization Strategy for Medical Diagnosis on Small Datasets.” Feb 2026.
(86.7% accuracy on UCI Cleveland Heart Disease; 93% model compression; ≈ 36 FLOPs/prediction.)
- **Martirosyan, V.** “Polynomial Regression Models for Interpretable Machine Learning: Theory, Optimization, and Medical Applications.” Feb 2026.
- **Martirosyan, V.** “Heart Disease Diagnosis Using Third-Order Polynomial Regression Models with L2 Regularization.” Feb 2026.
(81.8% accuracy, 10-fold CV; full interpretability via explicit polynomial coefficients.)
- **Martirosyan, V.** “Efficient Multi-Class ECG Classification Using Polynomial Regression with Analytical Solver.” *Applied Computational Intelligence and Soft Computing (major revision)*, 2026.
($82.75\% \pm 1.03\%$ six-class accuracy; 12 μ s inference; outperforms CNN, random forest, XGBoost, and SVM baselines.)
- **Martirosyan, V.** “Automated Multi-Class ECG Arrhythmia Classification Using an Ensemble of Polynomial Regression Models.” *IEEE ICCA 2026 (submitted)*, 2026.
(78.32% balanced accuracy; 123.1M parameters compressed 99.7% to 1.3 MB within ≈ 1 pp.)
- **Martirosyan, V.** “Explainable and Efficient Digit Classification via Polynomial Regression Networks: A Transparent Alternative to Deep Learning.” *arXiv preprint*, Jan 2026.
- **Martirosyan, V.** “Accuracy and Interpretability from a Single Solver: Polynomial Regression up to Degree 5 for Concrete Compressive Strength Prediction.” *ICACTE 2026 (published)*, 2026.
($R^2 = 0.859$, RMSE 6.25 MPa.)
- **Martirosyan, V.** “Efficient Crop Type Mapping via Degree-2 Polynomial Feature Expansion of Multi-Temporal Optical and SAR Data.” *Submitted to IEEE TGRS*, 2026.
(98.67% accuracy / macro-F1; $17\times$ SIMD speedup.)
- **Martirosyan, V.** “Sparse Polynomial Regression for Interpretable Yield-Strength Prediction in High-Entropy Alloys.” *IEEE ITT 2026 (under review)*, 2026.

($R^2 = 0.738$, tied with RF/XGBoost and above a neural network; 8/11 terms map to known hardening mechanisms.) Pending patent on a physics-descriptor design method.

Systems Verification & Software Engineering

- **Martirosyan, V.** et al. “Testing of Linux File System Drivers.” *Proceedings of ISP RAS*, vol. 23, Moscow, Dec 2012.
- **Martirosyan, V.** et al. “The Spruce System: Quality Verification of Linux File System Drivers.” *SYRCoSE 2012*, May 2012.
- **Martirosyan, V.** et al. “Dynamic Verification of Linux File System Drivers.” *CSIT 2011*, Sep 2011.
- **Martirosyan, V.** et al. “The Spruce System: Linux Kernel Module Verification Automation” (in Russian). *7th Annual Conference of RAU*, Dec 2012.
- **Martirosyan, V.** et al. “Fault Simulation in Linux Kernel Module Verification” (in Russian). *7th Annual Conference of RAU*, Dec 2012.
- **Martirosyan, V.** et al. “Linux Verification – Challenge of Today.” *7th Annual Conference of RAU*, Dec 2012.
- **Martirosyan, V.** et al. “X2C – XML To Code: Advantages of Standards.” *4th Scientific Annual Conference of RAU*, Dec 2009.
- **Martirosyan, V.** et al. “X2C: Code Generation Technology.”

Books & Tutorials

- **Martirosyan, V.** *Programming via Operating Systems, Second Edition*. Hilfmann Press, Sep 2024.
- **Martirosyan, V.** et al. *Tutorial: Basics of Programming in OS Linux Environment* (in Russian). MIPT, Apr 2011.
- **Martirosyan, V.** et al. *Tutorial: Programming in GNU/Linux Environment – From System Calls to the GUI* (in Russian). RAU Publishing, Dec 2010.

Translations

- *Jinnetic Engineering* by Richard Stallman, translated into Armenian. Dec 2010.

Technical Skills

- **Interpretable ML & Ocean Data:** Polynomial Regression Models (PRMs), stagewise/elastic-net regularization, feature importance; World Ocean Atlas 2018, IAP T/S Climatology 2024, EN4, global wave reanalysis; salinity/dissolved-oxygen/ wave-height reconstruction; profile-stratified, spatial-block, and temporal-holdout validation.
- **Edge / On-Device AI:** int8/int16 post-training quantization, fixed-point inference kernels, sub-MB models for microcontrollers and marine edge hardware (ESP32, RP2350, Arduino); latency and energy budgeting.
- **Machine Learning:** Supervised learning, regression modelling, interpretable model design, Python, PyTorch.
- **HPC & Parallelism:** LLVM/Clang (compiler pass development), binary lifting, OpenCL, MPI, POSIX Threads, auto-vectorisation, heterogeneous CPU-GPU execution, Xilinx ACAP.

- **Systems Programming:** C/C++ (expert), Linux kernel modules, device drivers (Ext4, XFS, JFS, Btrfs), WIN32 API, GTK+, IoT/embedded systems.
- **Web & Databases:** PHP/MySQL, ASP.NET/C#/MS SQL Server, SQL.
- **Tooling:** Git, Linux/Unix, L^AT_EX.
- **Languages:** Armenian (native), Russian (fluent), English (fluent).

Professional Development & Certifications

AI, Machine Learning & Data Science

- Fundamentals of Deep Learning *NVIDIA*, Apr 2025
- Artificial Intelligence Foundations: Thinking Machines *LinkedIn*, Jul 2024
- Introduction to Artificial Intelligence *LinkedIn*, Jul 2024
- Advanced NoSQL for Data Science *LinkedIn*, Jan 2025
- Statistics Foundations 1: The Basics *LinkedIn*, Sep 2024

Embedded Systems, FPGA & Hardware

- C Programming for Embedded Applications *LinkedIn*, Nov 2024
- Introduction to FreeRTOS and Basic Task Management *LinkedIn*, Nov 2024
- Getting Started with RISC-V *LinkedIn*, Jul 2024
- Learning Verilog for FPGA Development *LinkedIn*, Jul 2024
- Learning FPGA Development *LinkedIn*, Jul 2024
- Introduction to Quantum Computing *LinkedIn*, Jul 2024

Software Architecture & Engineering

- Software Architecture: Domain-Driven Design *LinkedIn*, Jul 2024
- Microservices Foundations *LinkedIn*, Jul 2024
- Microservices: Asynchronous Messaging *LinkedIn*, Jul 2024
- Microservices: Security *LinkedIn*, Jul 2024

Leadership & Communication

- How to Be an Effective Manager to Drive Impact *LinkedIn*, Jul 2024
- Storytelling to Influence Leadership and Decision Makers *LinkedIn*, May 2024
- Improving Your Thinking *LinkedIn*, Jul 2024
- Leading with Emotional Intelligence *LinkedIn*, Apr 2019
- Public Speaking Foundations *LinkedIn*, Feb 2019